

```
!pip install pandas numpy scikit-learn spacy transformers rapidfuzz python-pptx langdetect nltk
!python -m spacy download en_core_web_sm
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (4.57.3)
Requirement already satisfied: rapidfuzz in /usr/local/lib/python3.12/dist-packages (3.14.3)
Requirement already satisfied: python-pptx in /usr/local/lib/python3.12/dist-packages (1.0.2)
Requirement already satisfied: langdetect in /usr/local/lib/python3.12/dist-packages (1.0.9)
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.15)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.13)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.10)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.1.3)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.5.2)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.10)
Requirement already satisfied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.3)
Requirement already satisfied: typer-slim<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.21.1)
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (4.67.1)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.32.4)
Requirement already satisfied: pydantic!=1.8,!1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.1.6)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.1.6)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from spacy) (75.2.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (25.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.20.2)
Requirement already satisfied: huggingface-hub<1.0,>=0.34.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.4.3)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: Pillow>=3.2 in /usr/local/lib/python3.12/dist-packages (from python-pptx) (11.3.0)
Requirement already satisfied: XlsxWriter>=0.5.7 in /usr/local/lib/python3.12/dist-packages (from python-pptx) (3.2.9)
Requirement already satisfied: lxml>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from python-pptx) (6.0.2)
Requirement already satisfied: typing-extensions>=4.9.0 in /usr/local/lib/python3.12/dist-packages (from python-pptx) (4.1.1)
Requirement already satisfied: six in /usr/local/lib/python3.12/dist-packages (from langdetect) (1.17.0)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.1)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0,>=0.3) (2025.12.1)
Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0,>=0.3) (1.1.3)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!1.8) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!1.8) (2.41.4)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!1.8) (0.4.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0) (3.4.1)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0) (3.10.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0) (2.3.1)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0) (2025.11.12)
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->spacy) (1.3.0)
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->spacy) (0.0.1)
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2) (0.19.0)
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2) (7.0.5)
```

```
import pandas as pd
import numpy as np
import re
from sklearn.preprocessing import StandardScaler
from langdetect import detect
import nltk
nltk.download('punkt')
nltk.download('punkt_tab')
from nltk.tokenize import sent_tokenize, word_tokenize
import spacy
from rapidfuzz import process
from transformers import pipeline
from pptx import Presentation
from datetime import datetime
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!
```

#DATA INGESTION

```
df = pd.read_csv("/content/drive/MyDrive/Infosys_D/customer_support_tickets.csv")
print(df.shape)
df.head()
```

(8469, 17)

	Ticket ID	Customer Name	Customer Email	Customer Age	Customer Gender	Product Purchased	Date of Purchase	Ticket Type	Ticket Subject	Ticket Description
0	1	Marisa Obrien	carrollallison@example.com	32	Other	GoPro Hero	2021-03-22	Technical issue	Product setup	I'm having an issue with {product_purchased}
1	2	Jessica Rios	clarkeashley@example.com	42	Female	LG Smart TV	2021-05-22	Technical issue	Peripheral compatibility	I'm having an issue with {product_purchased}
2	3	Christopher Robbins	gonzalestracy@example.com	48	Other	Dell XPS	2020-07-14	Technical issue	Network problem	I'm facing a problem with {product_purchased}
3	4	Christina Dillon	bradleyolson@example.org	27	Female	Microsoft Office	2020-11-13	Billing inquiry	Account access	I'm having an issue with {product_purchased}
4	5	Alexander Carroll	bradleymark@example.com	67	Female	Autodesk AutoCAD	2020-02-04	Billing inquiry	Data loss	I'm having an issue with {product_purchased}

1 DATA CLEANING

#Missing Values

```
df.isnull().sum()
df.fillna("UNKNOWN", inplace=True)
```

```
/tmp/ipython-input-1979965605.py:4: FutureWarning: Setting an item of incompatible dtype is deprecated and will raise an error in the future.
df.fillna("UNKNOWN", inplace=True)
```

```
df.fillna("UNKNOWN", inplace=True)
```

```
# convert all columns to string type before fillna
df = df.astype("string") # uses pandas dedicated string dtype
df.fillna("UNKNOWN", inplace=True)
```

```
# only fill string columns with a string
for col in df.select_dtypes(include="object"):
    df[col].fillna("UNKNOWN", inplace=True)

# fill numeric columns with e.g. 0 or another numeric sentinel
for col in df.select_dtypes(include=["float64","int64"]):
    df[col].fillna(0, inplace=True)
```

```
df["Customer Name"] = df["Customer Name"].astype("string").fillna("UNKNOWN")
```

#Remove Duplicates

```
before = len(df)
df.drop_duplicates(inplace=True)
print("Removed:", before - len(df))
```

Removed: 0

#Fix Formatting

```
df.columns = df.columns.str.lower().str.replace(" ", "_")
df.head()
```

	ticket_id	customer_name	customer_email	customer_age	customer_gender	product_purchased	date_of_purchase	t
0	1	Marisa Obrien	carrollallison@example.com	32	Other	GoPro Hero	2021-03-22	
1	2	Jessica Rios	clarkeashley@example.com	42	Female	LG Smart TV	2021-05-22	
2	3	Christopher Robbins	gonzalestracy@example.com	48	Other	Dell XPS	2020-07-14	
3	4	Christina Dillon	bradleyolson@example.org	27	Female	Microsoft Office	2020-11-13	E
4	5	Alexander Carroll	bradleymark@example.com	67	Female	Autodesk AutoCAD	2020-02-04	E

2 DATA VALIDATION

```
assert "ticket_id" in df.columns
assert df["ticket_id"].notnull().all()
```

```
print(df.dtypes)
```

```
ticket_id          string[python]
customer_name      string[python]
customer_email     string[python]
customer_age       string[python]
customer_gender    string[python]
product_purchased  string[python]
date_of_purchase   string[python]
ticket_type        string[python]
ticket_subject     string[python]
ticket_description  string[python]
ticket_status      string[python]
resolution         string[python]
ticket_priority    string[python]
ticket_channel     string[python]
first_response_time string[python]
time_to_resolution string[python]
customer_satisfaction_rating string[python]
dtype: object
```

3 DATA TRANSFORMATION

```
if "created_at" in df.columns:
    df["created_at"] = pd.to_datetime(df["created_at"], errors="coerce")
```

```
scaler = StandardScaler()
if "priority" in df.columns:
    df["priority_scaled"] = scaler.fit_transform(df[["priority"]])
```

4 DATA FILTERING

```
df = df[df["ticket_description"] != "UNKNOWN"].copy()
df.head()
```

	ticket_id	customer_name	customer_email	customer_age	customer_gender	product_purchased	date_of_purchase	t
0	1	Marisa Obrien	carrollallison@example.com	32	Other	GoPro Hero	2021-03-22	
1	2	Jessica Rios	clarkeashley@example.com	42	Female	LG Smart TV	2021-05-22	
2	3	Christopher Robbins	gonzalestracy@example.com	48	Other	Dell XPS	2020-07-14	
3	4	Christina Dillon	bradleyolson@example.org	27	Female	Microsoft Office	2020-11-13	E
4	5	Alexander Carroll	bradleymark@example.com	67	Female	Autodesk AutoCAD	2020-02-04	E

5 DATA ENRICHMENT

```
df["text_length"] = df["ticket_description"].apply(len)
df["word_count"] = df["ticket_description"].apply(lambda x: len(x.split()))
df.head()
```

	ticket_id	customer_name	customer_email	customer_age	customer_gender	product_purchased	date_of_purchase	t
0	1	Marisa Obrien	carrollallison@example.com	32	Other	GoPro Hero	2021-03-22	
1	2	Jessica Rios	clarkeashley@example.com	42	Female	LG Smart TV	2021-05-22	
2	3	Christopher Robbins	gonzalestracy@example.com	48	Other	Dell XPS	2020-07-14	
3	4	Christina Dillon	bradleyolson@example.org	27	Female	Microsoft Office	2020-11-13	E
4	5	Alexander Carroll	bradleymark@example.com	67	Female	Autodesk AutoCAD	2020-02-04	E

6 DATA DEDUPLICATION

```
df = df.drop_duplicates(subset=["ticket_description"])
```

7 DATA MASKING

```
def mask_email(text):
    return re.sub(r'\S+@\S+', '[EMAIL]', text)
```

```
df["ticket_description"] = df["ticket_description"].apply(mask_email)
```

```
# df = df[df["ticket_status"] == "open"] # Commenting out this filter, as it was emptying the DataFrame
```

```
df["ticket_description"] = df["ticket_description"].apply(mask_email)
```

```
df.loc[:, "ticket_description"] = df["ticket_description"].apply(mask_email)
```

```
df["ticket_description"] = df["ticket_description"].apply(mask_email)
```

```
df = df.copy()
df.loc[:, "ticket_description"] = df["ticket_description"].apply(mask_email)
```

```
def mask_email(text):
    return re.sub(r'\S+@\S+', '[EMAIL]', text)
```

```
def mask_email(text):
    return re.sub(r'\S+@\S+', '[EMAIL]', text)

df["ticket_description"] = df["ticket_description"].apply(mask_email)
```

8 DATA STANDARDIZATION

```
df["ticket_status"] = df["ticket_status"].str.upper()
```

9 ERROR HANDLING & LOGGING

```
errors = []
for i,row in df.iterrows():
    try:
        len(row["description"])
    except:
        errors.append(i)
print("Error rows:", errors)

Error rows: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30]
```

10 METADATA HANDLING

```
df["ingested_at"] = datetime.now()
df["source"] = "Kaggle Customer Support Dataset"
```

11 SAMPLING

```
sample_df = df.sample(frac=0.2, random_state=42)
sample_df.head()
```

	ticket_id	customer_name	customer_email	customer_age	customer_gender	product_purchased	date_of_purchase
1653	1654	Alison Larsen	kneal@example.net	57	Female	Google Pixel	2021-06-16
7400	7401	Carrie Williams	joshua47@example.com	27	Male	Autodesk AutoCAD	2020-01-30
2198	2199	John Hale	moralesjeffrey@example.net	49	Male	iPhone	2020-08-11
6529	6530	Leslie Duffy	amydunn@example.net	27	Other	LG OLED	2021-11-05
1937	1938	Jennifer Mclean	holmesthomas@example.org	48	Male	Dell XPS	2021-09-04
5 rows × 21 columns							

#STAGE 1 - TEXT INGESTION

```
df["language"] = df["ticket_description"].apply(lambda x: detect(x) if x!="UNKNOWN" else "unknown")
df["sentences"] = df["ticket_description"].apply(sent_tokenize)
```

df.head()

	ticket_id	customer_name	customer_email	customer_age	customer_gender	product_purchased	date_of_purchase	t
0	1	Marisa Obrien	carrollallison@example.com	32	Other	GoPro Hero	2021-03-22	
1	2	Jessica Rios	clarkeashley@example.com	42	Female	LG Smart TV	2021-05-22	
2	3	Christopher Robbins	gonzalestracy@example.com	48	Other	Dell XPS	2020-07-14	
3	4	Christina Dillon	bradleyolson@example.org	27	Female	Microsoft Office	2020-11-13	E
4	5	Alexander Carroll	bradleymark@example.com	67	Female	Autodesk AutoCAD	2020-02-04	E

5 rows × 23 columns

df.describe()

	text_length	word_count	ingested_at
count	8077.000000	8077.000000	8077
mean	295.305311	47.436796	2026-01-21 06:59:58.440551
min	151.000000	21.000000	2026-01-21 06:59:58.440551
25%	278.000000	44.000000	2026-01-21 06:59:58.440551
50%	300.000000	49.000000	2026-01-21 06:59:58.440551
75%	319.000000	53.000000	2026-01-21 06:59:58.440551
max	397.000000	63.000000	2026-01-21 06:59:58.440551
std	36.086245	7.320994	NaN

df.info()

```
<class 'pandas.core.frame.DataFrame'>
Index: 8077 entries, 0 to 8468
Data columns (total 23 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   ticket_id                             8077 non-null   string
1   customer_name                         8077 non-null   string
2   customer_email                       8077 non-null   string
3   customer_age                         8077 non-null   string
4   customer_gender                      8077 non-null   string
5   product_purchased                   8077 non-null   string
6   date_of_purchase                    8077 non-null   string
7   ticket_type                         8077 non-null   string
8   ticket_subject                      8077 non-null   string
9   ticket_description                  8077 non-null   object
10  ticket_status                       8077 non-null   string
11  resolution                          8077 non-null   string
12  ticket_priority                     8077 non-null   string
13  ticket_channel                      8077 non-null   string
14  first_response_time                 8077 non-null   string
15  time_to_resolution                  8077 non-null   string
16  customer_satisfaction_rating        8077 non-null   string
17  text_length                         8077 non-null   int64
18  word_count                         8077 non-null   int64
19  ingested_at                        8077 non-null   datetime64[us]
20  source                             8077 non-null   object
21  language                           8077 non-null   object
22  sentences                           8077 non-null   object
dtypes: datetime64[us](1), int64(2), object(4), string(16)
memory usage: 1.5+ MB
```

```
nlp = spacy.load("en_core_web_sm")

def extract_entities(text):
    doc = nlp(text)
    return [(ent.text, ent.label_) for ent in doc.ents]

df["entities"] = df["ticket_description"].apply(extract_entities)
df[["ticket_description", "entities"]].head()
```

	ticket_description	entities
0	I'm having an issue with the {product_purchase...	[(71701, CARDINAL)]
1	I'm having an issue with the {product_purchase...	[]
2	I'm facing a problem with my {product_purchase...	[(yesterday, DATE)]
3	I'm having an issue with the {product_purchase...	[]
4	I'm having an issue with the {product_purchase...	[]

#STAGE 3 - ENTITY NORMALIZATION

```
canonical = {}

def normalize_entity(ent):
    if ent in canonical:
        return canonical[ent]
    best = process.extractOne(ent, canonical.keys())
    if best and best[1] > 85:
        canonical[ent] = canonical[best[0]]
    else:
        canonical[ent] = ent.lower()
    return canonical[ent]

df["normalized_entities"] = df["entities"].apply(
    lambda ents: [(normalize_entity(e),t) for e,t in ents]
)
```

#STAGE 4 - RELATION EXTRACTION (LLM Prompting)

```
classifier = pipeline("zero-shot-classification", model="facebook/bart-large-mnli")

relations = []
for text in df["ticket_description"].head(20):
    if len(text.split())>5:
        out = classifier(text, candidate_labels=["works_for", "uses", "located_in", "depends_on", "causes"])
        relations.append(out)
relations[:2]
```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF\_TOKEN` does not exist in your Colab secrets.

#TRIPLE CONSTRUCTION

Please note that authentication is recommended but still optional to access public models or datasets.

```
triples = []
for i,row in df.iterrows():
    for ent,typ in row["normalized_entities"]:
        triples.append((row["ticket_id"], ent, typ))
triples[:5]
```

```
[('cab.json', '1701', '899k', 'CARDINAL', '00:00:00', 18.6MB/s),
 ('3', 'yesterday', 'DATE'),
 ('merge.txt', 'yesterday', 'DATE', '00:00:00', 16.6MB/s),
 ('7', 'invalid credentials', 'WORK OF ART'),
 ('tokenizer.json', '1-800', 'CARDINAL', '00:00:00', 29.2MB/s)]
```

Device set to use cpu

#POST PROCESSING

steps mentioned in the user manual, but the issue persists.",

```
final_triples = list(set(triples))
len(final_triples)
```

```
0.2381872832775116,
54840.15540046989917755,
0.06569398939609528]],
```

```
df = pd.read_csv("/content/drive/MyDrive/Infosys_D/customer_support_tickets.csv")
print(df.shape)
df.head()
```

```
scores : [0.3433478165937805,
(84692303)0.356270313263,
0.21715635061264038,
Ticket ID, Customer Name, Customer Email, Customer Age, Customer Gender, Product Purchased, Date of Purchase, Ticket Type, Ticket Subject, Ticket Description
```

	Ticket ID	Customer Name	Customer Email	Customer Age	Customer Gender	Product Purchased	Date of Purchase	Ticket Type	Ticket Subject	Ticket Description
0	1	Marisa Obrien	carrollallison@example.com	32	Other	GoPro Hero	2021-03-22	Technical issue	Product setup	I'm having an issue with {product_purchased}
1	2	Jessica Rios	clarkeashley@example.com	42	Female	LG Smart TV	2021-05-22	Technical issue	Peripheral compatibility	I'm having an issue with {product_purchased}
2	3	Christopher Robbins	gonzalestracy@example.com	48	Other	Dell XPS	2020-07-14	Technical issue	Network problem	I'm facing a problem with {product_purchased}
3	4	Christina Dillon	bradleyolson@example.org	27	Female	Microsoft Office	2020-11-13	Billing inquiry	Account access	I'm having an issue with {product_purchased}
4	5	Alexander Carroll	bradleymark@example.com	67	Female	Autodesk AutoCAD	2020-02-04	Billing inquiry	Data loss	I'm having an issue with {product_purchased}

```
df.isnull().sum()
```

	0
Ticket ID	0
Customer Name	0
Customer Email	0
Customer Age	0
Customer Gender	0
Product Purchased	0
Date of Purchase	0
Ticket Type	0
Ticket Subject	0
Ticket Description	0
Ticket Status	0
Resolution	5700
Ticket Priority	0
Ticket Channel	0
First Response Time	2819
Time to Resolution	5700
Customer Satisfaction Rating	5700

dtype: int64

```
df.fillna("UNKNOWN", inplace=True)
```

```
/tmp/ipython-input-2372803106.py:1: FutureWarning: Setting an item of incompatible dtype is deprecated and will raise an error in the future.
df.fillna("UNKNOWN", inplace=True)
```

```
df.columns
```

```
Index(['Ticket ID', 'Customer Name', 'Customer Email', 'Customer Age',
      'Customer Gender', 'Product Purchased', 'Date of Purchase',
      'Ticket Type', 'Ticket Subject', 'Ticket Description', 'Ticket Status',
      'Resolution', 'Ticket Priority', 'Ticket Channel',
      'First Response Time', 'Time to Resolution',
      'Customer Satisfaction Rating'],
      dtype='object')
```

```
df.columns = df.columns.str.lower().str.replace(" ", "_")
df = df[df["ticket_description"] != "UNKNOWN"]
```

```
print(df.shape)
```

```
(8469, 17)
```

```
print("Rows:", len(df))
df.sample(5)
```

Rows: 8469

	ticket_id	customer_name	customer_email	customer_age	customer_gender	product_purchased	date_of_purchas
7619	7620	Anne Deleon	michaelfrazier@example.com	52	Other	Bose SoundLink Speaker	2020-04-0
5833	5834	Jake Murray	robert86@example.com	45	Female	Google Pixel	2021-12-2
3373	3374	Sherry Olsen	dawn90@example.com	54	Other	Amazon Echo	2020-08-1
6738	6739	Christopher Ellis	christinerichards@example.net	69	Male	Lenovo ThinkPad	2021-05-1
2481	2482	Michael Kelly	morrisjeff@example.com	56	Other	Samsung Soundbar	2021-03-0

Start coding or [generate](#) with AI.

```
print(df.columns)
print(df.info())
print(df.head(3))

Index(['ticket_id', 'customer_name', 'customer_email', 'customer_age',
      'customer_gender', 'product_purchased', 'date_of_purchase',
      'ticket_type', 'ticket_subject', 'ticket_description', 'ticket_status',
      'resolution', 'ticket_priority', 'ticket_channel',
      'first_response_time', 'time_to_resolution',
      'customer_satisfaction_rating'],
      dtype='object')
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8469 entries, 0 to 8468
Data columns (total 17 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   ticket_id                            8469 non-null   int64
1   customer_name                        8469 non-null   object
2   customer_email                      8469 non-null   object
3   customer_age                        8469 non-null   int64
4   customer_gender                     8469 non-null   object
5   product_purchased                   8469 non-null   object
6   date_of_purchase                    8469 non-null   object
7   ticket_type                         8469 non-null   object
8   ticket_subject                      8469 non-null   object
9   ticket_description                  8469 non-null   object
10  ticket_status                       8469 non-null   object
11  resolution                          8469 non-null   object
12  ticket_priority                     8469 non-null   object
13  ticket_channel                      8469 non-null   object
14  first_response_time                 8469 non-null   object
15  time_to_resolution                  8469 non-null   object
16  customer_satisfaction_rating        8469 non-null   object
dtypes: int64(2), object(15)
memory usage: 1.1+ MB
None
      ticket_id      customer_name      customer_email      customer_age \
0              1      Marisa Obrien  carrollallison@example.com      32
1              2      Jessica Rios    clarkeashley@example.com      42
2              3  Christopher Robbins  gonzalestracy@example.com      48

      customer_gender  product_purchased  date_of_purchase      ticket_type \
0              Other      GoPro Hero    2021-03-22  Technical issue
1              Female      LG Smart TV    2021-05-22  Technical issue
2              Other      Dell XPS    2020-07-14  Technical issue

      ticket_subject \
0      Product setup
1  Peripheral compatibility
2      Network problem

      ticket_description \
0  I'm having an issue with the {product_purchase...
1  I'm having an issue with the {product_purchase...
2  I'm facing a problem with my {product_purchase...

      ticket_status      resolution \
```

0	Pending Customer Response	UNKNOWN
1	Pending Customer Response	UNKNOWN
2	Closed	Case maybe show recently my computer follow.

```
import networkx as nx

G = nx.DiGraph()

for _, r in df.iterrows():
    ticket = f"Ticket:{r['ticket_id']}"
    customer = f"Customer:{r['customer_name']}" # Changed from customer_id to customer_name
    # No direct 'agent_id' column, skipping for now or default to 'UNKNOWN_AGENT'
    agent = f"Agent:UNKNOWN_AGENT" # Placeholder as 'agent_id' column is not in df
    issue = f"Issue:{r['ticket_subject']}" # Changed from issue_type to ticket_subject
    product = f"Product:{r['product_purchased']}" # Changed from product to product_purchased

    G.add_node(ticket, type="Ticket")
    G.add_node(customer, type="Customer")
    G.add_node(agent, type="Agent")
    G.add_node(issue, type="Issue")
    G.add_node(product, type="Product")

    G.add_edge(customer, ticket, relation="CREATED")
    G.add_edge(agent, ticket, relation="HANDLED")
    G.add_edge(ticket, issue, relation="ABOUT")
    G.add_edge(ticket, product, relation="FOR_PRODUCT")
```

```
nx.write_gml(G, "customer_support_kg.gml")
```

```
edges = []
for u, v, d in G.edges(data=True):
    edges.append([u, d["relation"], v])

import pandas as pd
pd.DataFrame(edges, columns=["subject", "predicate", "object"]).to_csv("triples.csv", index=False)
```

```
import pandas as pd

df = pd.read_csv("/content/drive/MyDrive/Infosys_D/customer_support_tickets.csv")
print("Shape:", df.shape)
print(df.head())
```

```
Shape: (8469, 17)
```

	Ticket ID	Customer Name	Customer Email	Customer Age	\
0	1	Marisa Obrien	carrollallison@example.com	32	
1	2	Jessica Rios	clarkeashley@example.com	42	
2	3	Christopher Robbins	gonzalestracy@example.com	48	
3	4	Christina Dillon	bradleyolson@example.org	27	
4	5	Alexander Carroll	bradleymark@example.com	67	

	Customer	Gender	Product Purchased	Date of Purchase	Ticket Type	\
0	Other		GoPro Hero	2021-03-22	Technical issue	
1	Female		LG Smart TV	2021-05-22	Technical issue	
2	Other		Dell XPS	2020-07-14	Technical issue	
3	Female		Microsoft Office	2020-11-13	Billing inquiry	
4	Female		Autodesk AutoCAD	2020-02-04	Billing inquiry	

	Ticket Subject	\
0	Product setup	
1	Peripheral compatibility	
2	Network problem	
3	Account access	
4	Data loss	

	Ticket Description	\
0	I'm having an issue with the {product_purchase...}	
1	I'm having an issue with the {product_purchase...}	
2	I'm facing a problem with my {product_purchase...}	
3	I'm having an issue with the {product_purchase...}	
4	I'm having an issue with the {product_purchase...}	

	Ticket Status	Resolution	\
0	Pending Customer Response	NaN	
1	Pending Customer Response	NaN	
2	Closed	Case maybe show recently my computer follow.	
3	Closed	Try capital clearly never color toward story.	
4	Closed	West decision evidence bit.	

	Ticket Priority	Ticket Channel	First Response Time	Time to Resolution	\
0	Critical	Social media	2023-06-01 12:15:36	NaN	
1	Critical	Chat	2023-06-01 16:45:38	NaN	
2	Low	Social media	2023-06-01 11:14:38	2023-06-01 18:05:38	
3	Low	Social media	2023-06-01 07:29:40	2023-06-01 01:57:40	

4 Low Email 2023-06-01 00:12:42 2023-06-01 19:53:42

Customer Satisfaction Rating

0 NaN
1 NaN
2 3.0
3 3.0
4 1.0

display(df.head())

	Ticket ID	Customer Name	Customer Email	Customer Age	Customer Gender	Product Purchased	Date of Purchase	Ticket Type	Ticket Subject	Ticket Description
0	1	Marisa Obrien	carrollallison@example.com	32	Other	GoPro Hero	2021-03-22	Technical issue	Product setup	I'm having an issue with {product_purchased}
1	2	Jessica Rios	clarkeashley@example.com	42	Female	LG Smart TV	2021-05-22	Technical issue	Peripheral compatibility	I'm having an issue with {product_purchased}
2	3	Christopher Robbins	gonzalestracy@example.com	48	Other	Dell XPS	2020-07-14	Technical issue	Network problem	I'm facing a problem with {product_purchased}
3	4	Christina Dillon	bradleyolson@example.org	27	Female	Microsoft Office	2020-11-13	Billing inquiry	Account access	I'm having an issue with {product_purchased}
4	5	Alexander Carroll	bradleymark@example.com	67	Female	Autodesk AutoCAD	2020-02-04	Billing inquiry	Data loss	I'm having an issue with {product_purchased}

```
import pandas as pd
import networkx as nx

df = pd.read_csv("/content/drive/MyDrive/Infosys_D/customer_support_tickets.csv")

# Standardize column names
df.columns = df.columns.str.lower().str.replace(" ", "_")

G = nx.DiGraph()

for _, r in df.iterrows():
    ticket = f"Ticket:{r['ticket_id']}"
    customer = f"Customer:{r['customer_name']}" # Use 'customer_name' instead of 'customer_id'
    agent = f"Agent:UNKNOWN_AGENT" # No 'agent_id' column, using a placeholder
    issue = f"Issue:{r['ticket_subject']}" # Use 'ticket_subject' instead of 'issue_type'
    product = f"Product:{r['product_purchased']}" # Use 'product_purchased' instead of 'product'

    G.add_node(ticket, type="Ticket")
    G.add_node(customer, type="Customer")
    G.add_node(agent, type="Agent")
    G.add_node(issue, type="Issue")
    G.add_node(product, type="Product")

    G.add_edge(customer, ticket, relation="CREATED")
    G.add_edge(agent, ticket, relation="HANDLED")
    G.add_edge(ticket, issue, relation="ABOUT")
    G.add_edge(ticket, product, relation="FOR_PRODUCT")
```

```
print("Nodes:", G.number_of_nodes())
print("Edges:", G.number_of_edges())
```

Nodes: 16556
Edges: 33876

```
for u, v, d in list(G.edges(data=True))[:10]:
    print(u, "--", d["relation"], "-->", v)
```

```
Ticket:1 -- ABOUT --> Issue:Product setup
Ticket:1 -- FOR_PRODUCT --> Product:GoPro Hero
Customer:Marisa Obrien -- CREATED --> Ticket:1
Agent:UNKNOWN_AGENT -- HANDLED --> Ticket:1
Agent:UNKNOWN_AGENT -- HANDLED --> Ticket:2
```

```
Agent:UNKNOWN_AGENT -- HANDLED --> Ticket:3
Agent:UNKNOWN_AGENT -- HANDLED --> Ticket:4
Agent:UNKNOWN_AGENT -- HANDLED --> Ticket:5
Agent:UNKNOWN_AGENT -- HANDLED --> Ticket:6
Agent:UNKNOWN_AGENT -- HANDLED --> Ticket:7
```

```
nx.write_gml(G, "customer_support_kg.gml")
print("Knowledge graph saved as customer_support_kg.gml")
```

Knowledge graph saved as customer_support_kg.gml

```
print("Nodes:", G.number_of_nodes())
print("Edges:", G.number_of_edges())

# Count node types
from collections import Counter
types = Counter([d.get("type") for _, d in G.nodes(data=True)])
print(types)
```

```

NameError                                Traceback (most recent call last)
/tmp/ipython-input-2387389180.py in <cell line: 0>()
----> 1 print("Nodes:", G.number_of_nodes())
      2 print("Edges:", G.number_of_edges())
      3
      4 # Count node types
      5 from collections import Counter

NameError: name 'G' is not defined

```

```
import pandas as pd
import networkx as nx

# Load your CSV again
df = pd.read_csv("/content/drive/MyDrive/Infosys_D/customer_support_tickets.csv")

# Standardize column names
df.columns = df.columns.str.lower().str.replace(" ", "_")

# Build Knowledge Graph
G = nx.DiGraph()

for _, r in df.iterrows():
    # Create nodes
    n1 = f"Ticket {r['Ticket ID']}"
```